

LINUX

Aditi Biju

261100690027

THEORY ASSIGNMENT 2

Compare Permissions on Regular Files and Directories. Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

1. Setting up

```
(aduuu@kali)-[~]
$ mkdir sample_dir1
(aduuu@kali)-[~]
$ touch sample.txt
(aduuu@kali)-[~]
$ touch sample_dir/inside.txt
(aduuu@kali)-[~]
$ echo 'hello, this is sample.txt' > sample.txt
$ echo "setup" > sample.txt
$ echo 'hello, this is sample.txt' > sample.txt
$ echo "setup"^C
(aduuu@kali)-[~]
$ ls -l sample.txt
-rw-rw-r-- 1 aduuu aduuu 0 Aug 22 20:07 sample.txt
(aduuu@kali)-[~]
$ ls -l sample_dir
total 0
-rw-rw-r-- 1 aduuu aduuu 0 Aug 22 20:07 inside.txt
(aduuu@kali)-[~]
$ ls -l sample_dir
total 0
-rw-rw-r-- 1 aduuu aduuu 0 Aug 22 20:07 inside.txt
```

2. Permission 400 — r--

```
(aduuu@kali)-[~]
$ chmod 400 sample.txt
(aduuu@kali)-[~]
$ chmod 400 sample_dir
(aduuu@kali)-[~]
$ echo "400 (r-) "
400 (r-)
(aduuu@kali)-[~]
$ echo "read file"
read file
(aduuu@kali)-[~]
$ cat sample.txt
(aduuu@kali)-[~]
$ echo "modify file"
modify file
(aduuu@kali)-[~]
$ echo "trying to modify" >> sample.txt
bash: sample.txt: Permission denied
(aduuu@kali)-[~]
$ echo "list directory"
list directory
(aduuu@kali)-[~]
$ ls sample_dir
ls: cannot access 'sample_dir/inside.txt': Permission denied
inside.txt
(aduuu@kali)-[~]
$ ls sample_dir1
(aduuu@kali)-[~]
$ chmod 400 sample_dir1
(aduuu@kali)-[~]
$ echo "enter directory"
enter directory
(aduuu@kali)-[~]
$ cd sample_dir1
bash: cd: sample_dir1: Permission denied
```

File can be read but not modified. Directory cannot be listed or entered.

3. Permission 200 — -w-

```
(aduuu@kali)-[~]
$ chmod 200 sample.txt
(aduuu@kali)-[~]
$ chmod 200 sample_dir1
(aduuu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied
(aduuu@kali)-[~]
$ echo "modified" >> sample.txt
(aduuu@kali)-[~]
$ ls sample_dir1
ls: cannot open directory 'sample_dir1': Permission denied
(aduuu@kali)-[~]
$ cd sample_dir1
bash: cd: sample_dir1: Permission denied
(aduuu@kali)-[~]
$ touch sample_dir1/newfile.txt
touch: cannot touch 'sample_dir1/newfile.txt': Permission denied
(aduuu@kali)-[~]
$
```

Write permission alone on a directory is not enough to access its contents.

4. Permission 100 — --x

```
(aduuu@kali)-[~]
$ chmod 100 sample.txt
(aduuu@kali)-[~]
$ chmod 100 sample_dir1
(aduuu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied
(aduuu@kali)-[~]
$ ./sample.txt
/bin/bash: ./sample.txt: Permission denied
(aduuu@kali)-[~]
$ ls sample_dir1
ls: cannot open directory 'sample_dir1': Permission denied
(aduuu@kali)-[~]
$ cd sample_dir1
(aduuu@kali)-[~/sample_dir1]
$
```

Execute permission on a directory allows traversal/entering, but without r, its contents cannot be listed.

5. Permission 500 — r-x

```
(aduuu@kali)-[~]
$ chmod 500 sample.txt
(aduuu@kali)-[~]
$ chmod 500 sample_dir1
(aduuu@kali)-[~]
$ cat sample.txt
modified text
modified again
final modification
modified
(aduuu@kali)-[~]
$ echo "modified" >> sample.txt
bash: sample.txt: Permission denied
(aduuu@kali)-[~]
$ ls sample_dir1
created.txt
(aduuu@kali)-[~]
$ cd sample_dir1
(aduuu@kali)-[~/sample_dir1]
$
```

The file can be read but not modified. The directory can be entered and listed, but files cannot be created or deleted.

6. Permission 600 — rw-

```
(aduuu@kali)-[~]
$ chmod 600 sample.txt
(aduuu@kali)-[~]
$ chmod 600 sample_dir1
(aduuu@kali)-[~]
$ cat sample.txt
modified text
modified again
final modofication
modified
(aduuu@kali)-[~]
$ echo "modified" >> sample.txt
(aduuu@kali)-[~]
$ ls sample_dir1
ls: cannot access 'sample_dir1/created.txt': Permission denied
created.txt
(aduuu@kali)-[~]
$ cd sample_dir1
bash: cd: sample_dir1: Permission denied
(aduuu@kali)-[~]
$ touch sample_dir1/newfile.txt
touch: cannot touch 'sample_dir1/newfile.txt': Permission denied
```

For the file, read and write work normally. For the directory, w without x does not provide useful access to the directory's contents.

7. Permission 700 — rwx

```
(aduuu@kali)-[~]
$ chmod 700 sample.txt
(aduuu@kali)-[~]
$ chmod 700 sample_dir1
(aduuu@kali)-[~]
$ cat sample.txt
modified text
modified again
final modofication
modified
modified
(aduuu@kali)-[~]
$ echo "modified" >> sample.txt
(aduuu@kali)-[~]
$ ./sample.txt
./sample.txt: line 1: modified: command not found
./sample.txt: line 2: modified: command not found
./sample.txt: line 3: final: command not found
./sample.txt: line 4: modified: command not found
./sample.txt: line 5: modified: command not found
./sample.txt: line 6: modified: command not found
(aduuu@kali)-[~]
$ ls sample_dir1
created.txt
(aduuu@kali)-[~]
$ cd sample_dir1
(aduuu@kali)-[~/sample_dir1]
$ touch newfile.txt
(aduuu@kali)-[~/sample_dir1]
$ rm newfile.txt
(aduuu@kali)-[~/sample_dir1]
$
```

The owner has full permissions on both the file and directory.

Permission	Regular File (sample.txt) – Observation	Directory (sample_dir) – Observation
400 (r--)	The file can be opened and read but cannot change its contents	The directory could not be entered.
200 (-w-)	The file cannot be read, but its contents can be changed.	The directory cannot be viewed, entered, or used to add a file.
100 (--x)	The file cannot be read or run in this test.	The directory can be entered, but its contents cannot be listed.
500 (r-x)	The file can be read, but changes are not permitted.	The directory can be viewed and entered, but files cannot be created or removed.
600 (rw-)	Both reading and changing the file are permitted.	The directory can be listed, but it cannot be entered or used to create files.
700 (rwx)	The owner can read and change the file and has execute permission.	The directory can be listed, entered, and files can be created or removed.

Answers to the Questions

1. **What does read (r) permission allow you to do with a regular file?**

It gives the user access to open the file and see the information stored in it.

2. **What does read (r) permission allow you to do with a directory?**

It allow the user to see the list of files and folders contained within the directory.

3. **What does write (w) permission mean for a regular file?**

It gives permission to make changes to the data stored in the file.

4. **What does write (w) permission mean for a directory?**

It allows you to create, delete, or rename files and folders inside the directory. The x permission is also needed to access them.

5. **What does execute (x) permission mean for a regular file?**

It allows the file to be run as a program or script when the file is suitable for execution.

6. **Why is execute (x) permission important for accessing a directory?**

It provides the ability to enter and move through the directory and access items stored inside it.

7. **Which permission combination allows the owner to fully access a regular file?**

700 (rwx) gives the owner all three permissions: reading, writing, and executing the file.

8. **Which permission combination allows the owner to fully access a directory?**

700 (rwx) provides complete access to the directory, including listing, entering, creating, and removing entries.

9. **Why do the same r, w, and x permissions behave differently for a regular file and a directory?**

The same permissions work differently because a file and a directory are different.

For a file, they control reading, changing, and running it. For a directory, they control viewing, changing, and entering it.